

Predicting Diabetes Risk from Clinical and Demographic Data: A Comparative Machine Learning Study

CBIO313: Machine Learning & Data Mining
Project Report

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1. Abstract

This project investigates the predictive power of clinical and demographic variables — most notably HbA1c level and blood glucose level — in identifying diabetes using machine learning. Working from a health dataset of 100,000 individuals, six supervised classification models were trained and compared: Random Forest, XGBoost, Logistic Regression, K-Nearest Neighbors, Decision Tree, and Naive Bayes. Model performance was evaluated using accuracy, precision, recall, F1 score, confusion matrices, and ROC-AUC. XGBoost and Random Forest achieved the strongest overall balance between precision and recall (F1 = 0.81 and 0.80 respectively), while feature importance analysis confirmed HbA1c level and blood glucose level as the two dominant predictors, together accounting for roughly 70% of the models' predictive power.

2. Introduction

Diabetes is a chronic metabolic disorder affecting millions of people worldwide, with serious long-term complications including cardiovascular disease, kidney failure, nerve damage, and vision loss if left undiagnosed or poorly managed. Early and accurate diagnosis is therefore critical. Hemoglobin A1c (HbA1c) is a widely used clinical biomarker that reflects average blood glucose levels over two to three months, offering a more stable measure of glycemic control than a single glucose reading.

This study explores how well HbA1c level, together with other demographic and clinical features such as age, BMI, blood glucose level, smoking history, hypertension, and heart disease, can predict diabetes status using supervised machine learning. Several classification algorithms — ranging from simple linear models to ensemble methods — were trained and compared to identify which approach offers the most reliable predictions.

3. Methodology

3.1 Dataset

The dataset consists of 100,000 patient records with 16 columns, including age, gender, BMI, HbA1c level, blood glucose level, smoking history, hypertension, heart disease, and the binary target variable diabetes (0 = No, 1 = Yes). The dataset is highly imbalanced: of the 19,994 records in the held-out test split, 18,291 (~91.5%) are non-diabetic and 1,703 (~8.5%) are diabetic.

3.2 Data Preprocessing

- Checked the dataset for missing values (none found) and duplicate rows (14 duplicate records identified).
- Encoded gender (Male = 0, Female = 1) and smoking_history (No Info = 0, never = 1, former = 2, current = 3, not current = 4, ever = 5) as numeric categories.
- Applied StandardScaler to age, bmi, hbA1c_level, and blood_glucose_level.
- Split the data 80% training / 20% testing.

3.3 Exploratory Data Analysis

Exploratory analysis was carried out to understand the distribution of the target variable and the relationship between key clinical features and diabetes status.

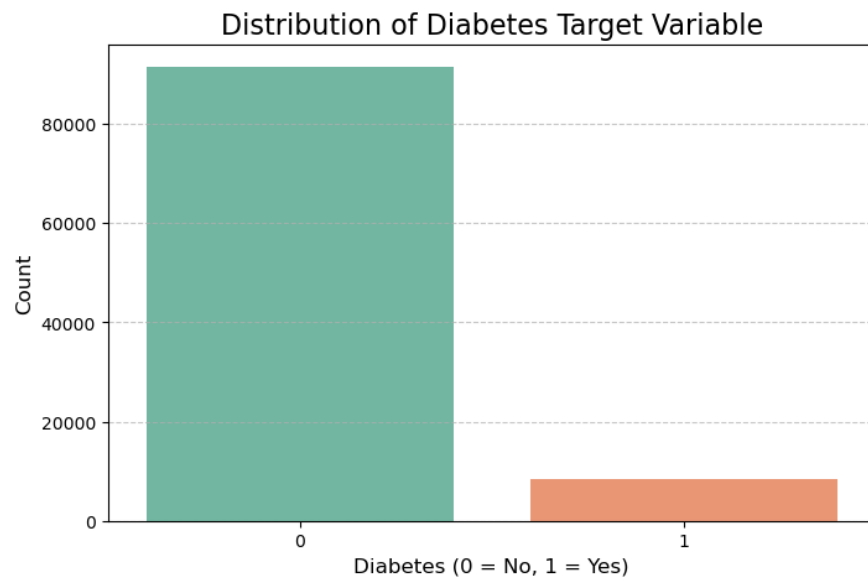


Fig. 1 — Distribution of the diabetes target variable, confirming the ~91.5% / 8.5% class imbalance.

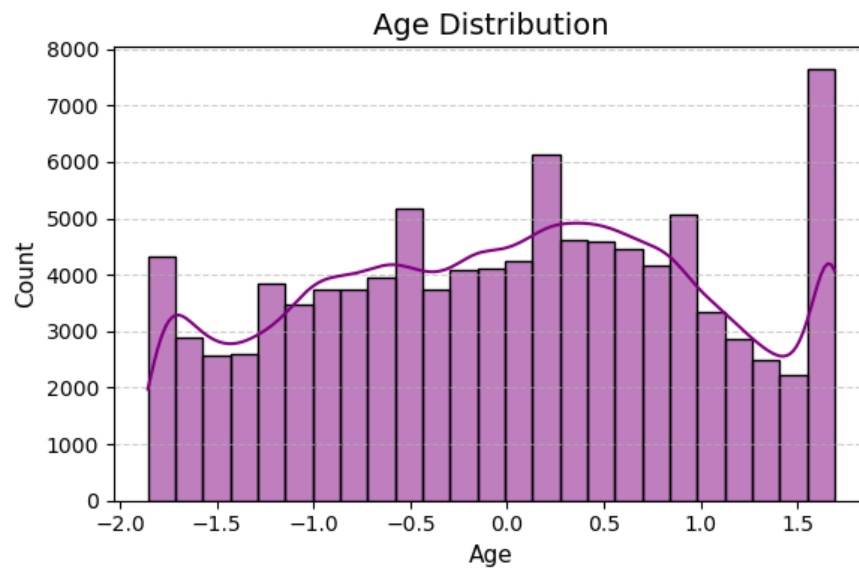


Fig. 2 — Age distribution across the dataset.

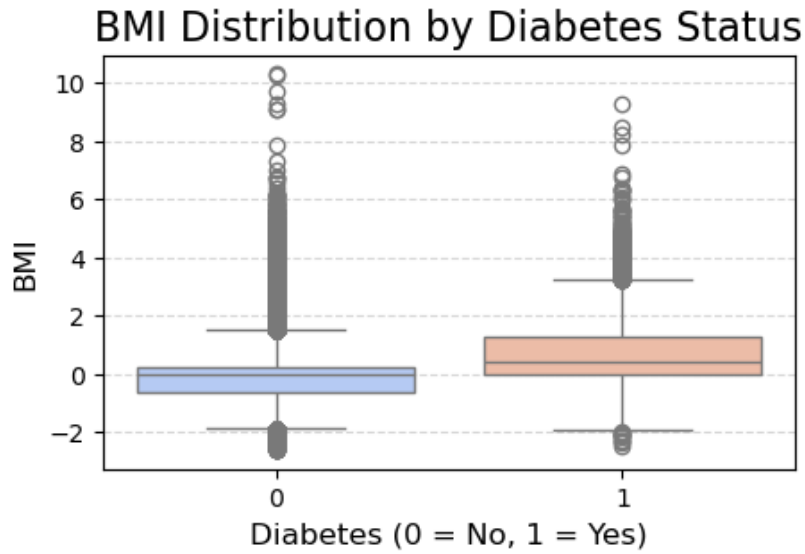


Fig. 3 — BMI distribution by diabetes status; diabetic patients tend toward higher BMI.

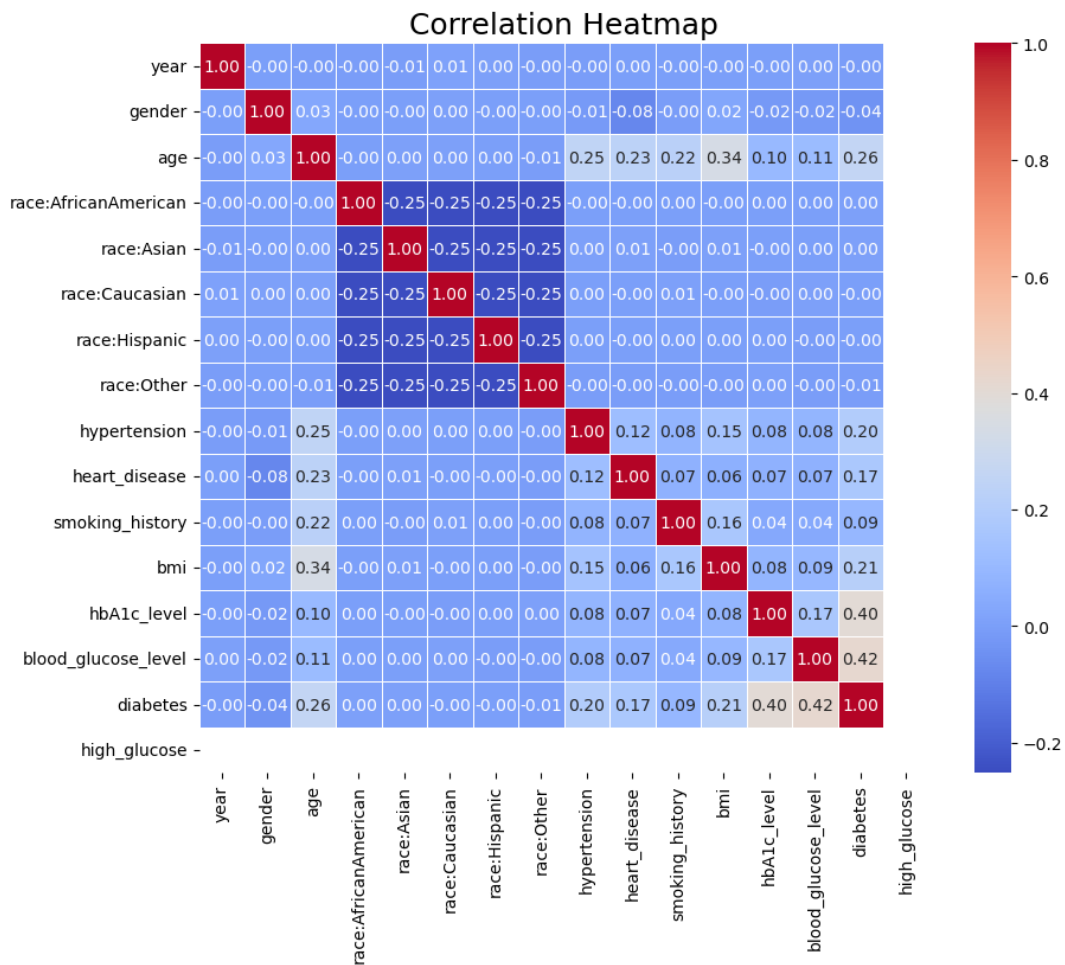


Fig. 4 — Correlation heatmap across all numeric and encoded features.

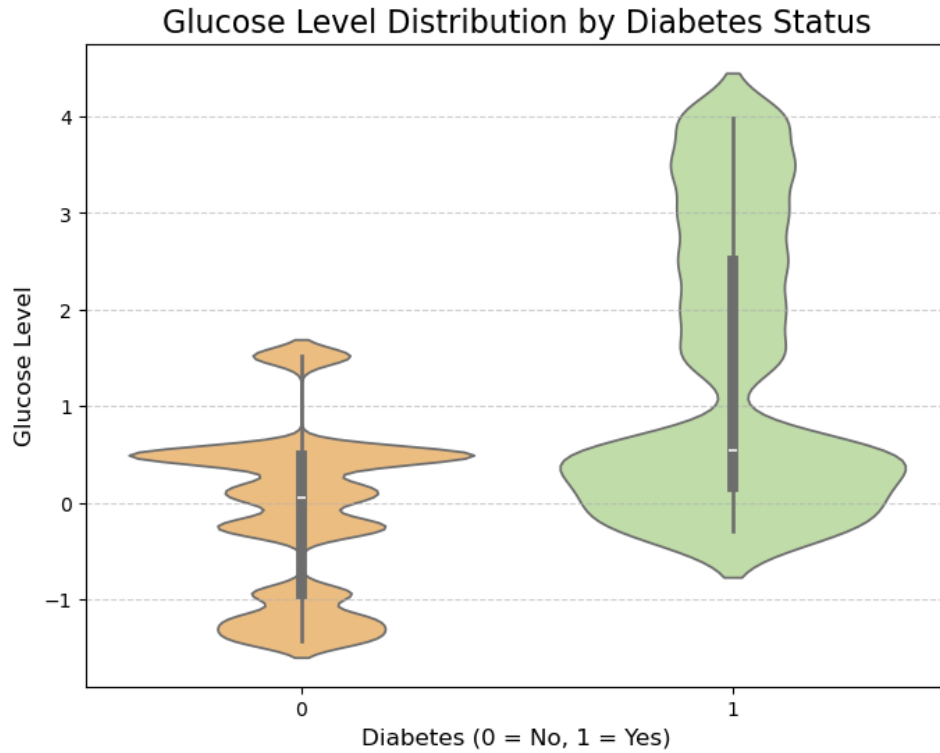


Fig. 5 — Blood glucose level by diabetes status; diabetic patients show a clear rightward shift toward higher glucose.

3.4 Feature Engineering & Selection

A new binary feature, `high_glucose`, was engineered (1 if `blood_glucose_level` > 140, indicating high risk, else 0). Feature selection used an embedded method: Random Forest feature importances were computed and the top 5 features were retained for final model training.

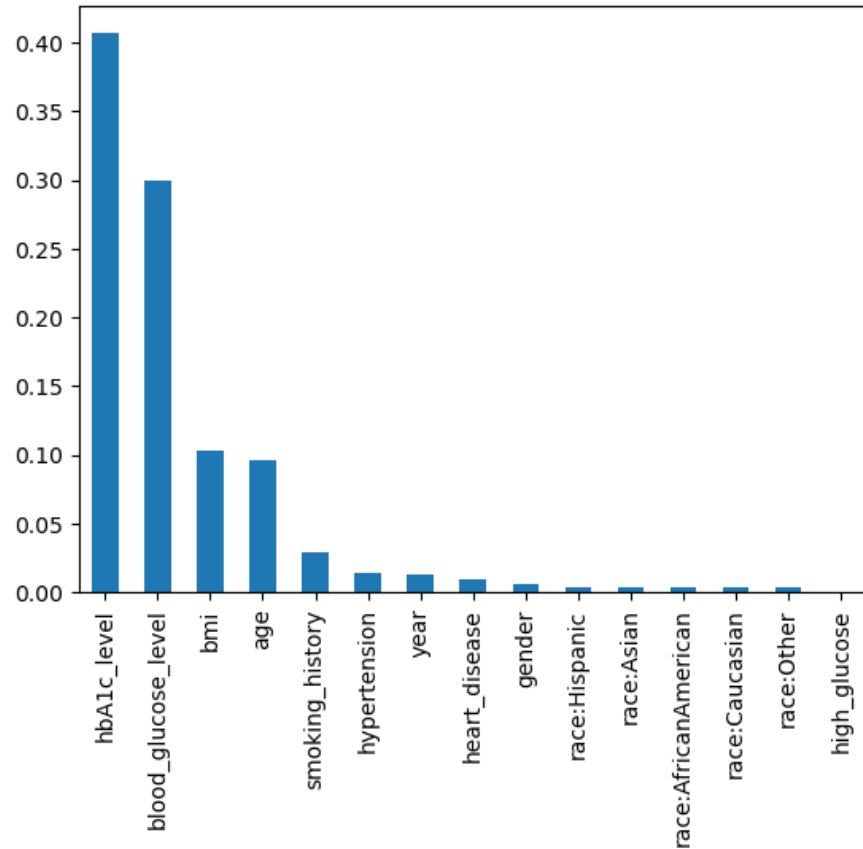


Fig. 6 — Random Forest feature importances. HbA1c level and blood glucose level dominate.

Feature	Importance
hbA1c_level	0.390
blood_glucose_level	0.317
bmi	0.104
age	0.094
smoking_history	0.029
hypertension	0.014
year	0.013
heart_disease	0.011
gender	0.007

The top 5 selected features were: **hbA1c_level**, **blood_glucose_level**, **bmi**, **age**, and **smoking_history**.

3.5 Linearity Checks

Logistic-regression trend plots were generated for each of the top 5 selected features to confirm a monotonic relationship with the diabetes target before finalizing the feature set.

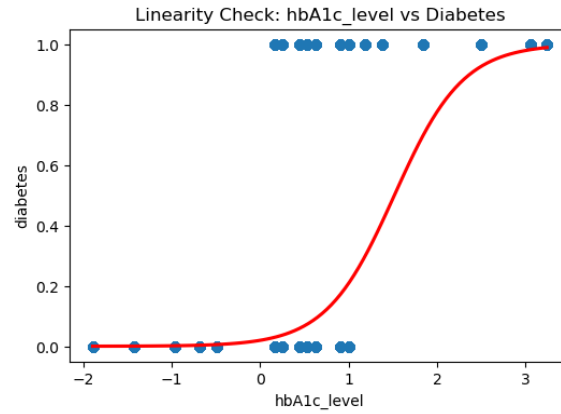


Fig. 7a — Linearity check: *hbA1c_level* vs. *diabetes*.

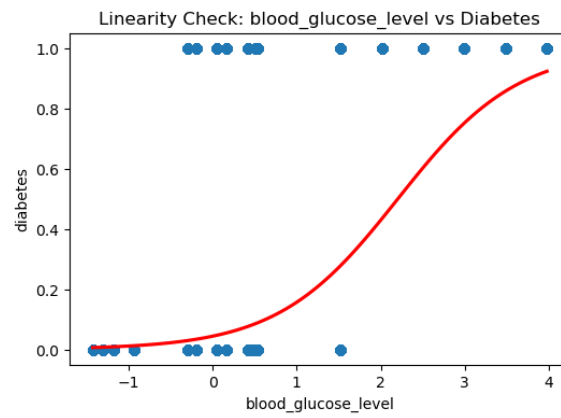


Fig. 7b — Linearity check: *blood_glucose_level* vs. *diabetes*.

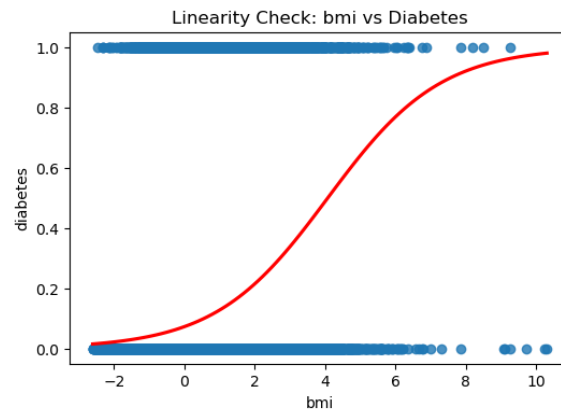


Fig. 7c — Linearity check: *bmi* vs. *diabetes*.

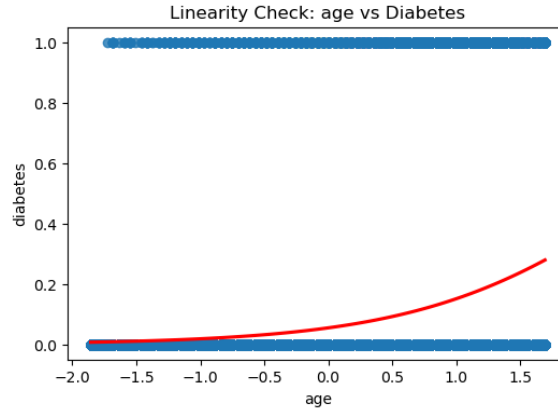


Fig. 7d — Linearity check: age vs. diabetes.

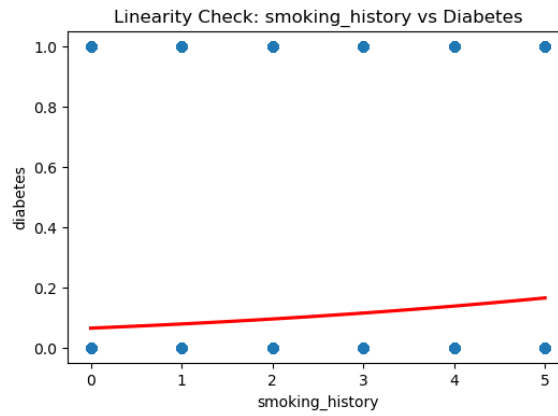


Fig. 7e — Linearity check: smoking_history vs. diabetes.

3.6 Models Trained

Six supervised classification models were trained and evaluated on the selected features: Random Forest, XGBoost, Logistic Regression, K-Nearest Neighbors (KNN), Decision Tree, and Naive Bayes. Each model was assessed using accuracy, precision, recall, F1 score, a confusion matrix, and an ROC curve with AUC score.

4. Results & Discussion

Table 1 summarizes the performance of all six models on the 19,994-record test set.

Model	Accuracy	Precision	Recall	F1 Score	AUC
Random Forest	0.97	0.98	0.68	0.80	0.96
XGBoost	0.97	0.96	0.69	0.81	0.98
Logistic Regression	0.96	0.87	0.62	0.73	0.96
K-Nearest Neighbors	0.96	0.90	0.57	0.70	0.89
Decision Tree	0.95	0.70	0.73	0.71	0.85

Model	Accuracy	Precision	Recall	F1 Score	AUC
Naive Bayes	0.91	0.46	0.66	0.54	0.92

4.1 Random Forest

Random Forest achieved 0.97 accuracy, 0.98 precision, 0.68 recall, and 0.80 F1 score. The confusion matrix shows 18,264 true negatives and 1,158 true positives with only 27 false positives (AUC = 0.96), reflecting very high specificity.

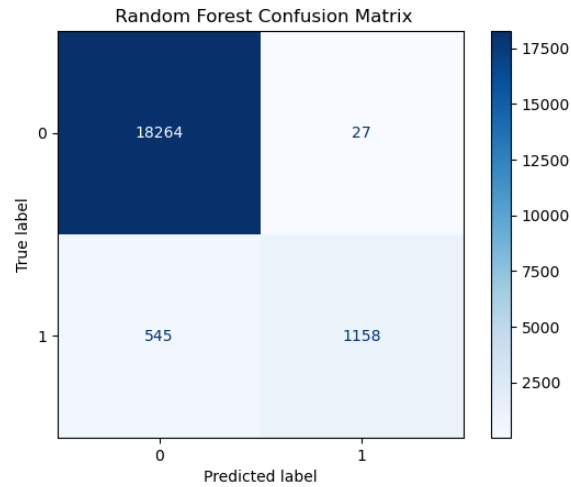


Fig. 8a — Random Forest confusion matrix.

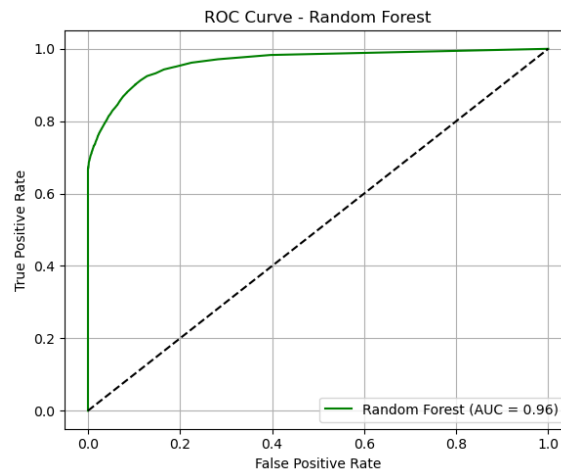


Fig. 8b — Random Forest ROC curve (AUC = 0.96).

4.2 XGBoost

XGBoost achieved 0.97 accuracy, 0.96 precision, 0.69 recall, and the highest F1 score of 0.81 among all models tested, with an AUC of 0.98. With 18,247 true negatives and 1,180 true positives, it offers the best balance between catching diabetic cases and avoiding false alarms.

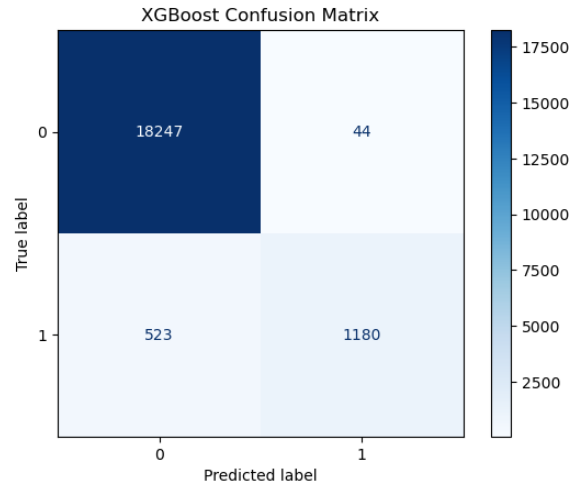


Fig. 9a — XGBoost confusion matrix.

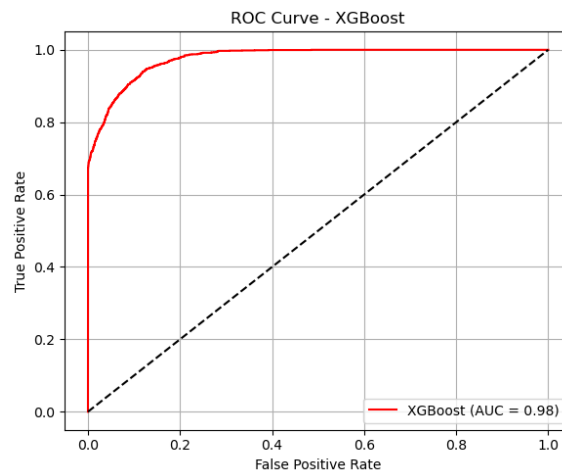


Fig. 9b — XGBoost ROC curve ($AUC = 0.98$).

4.3 Logistic Regression

Logistic Regression achieved 0.96 accuracy, 0.87 precision, 0.62 recall, and 0.73 F1 score ($AUC = 0.96$). It offers strong interpretability but captures nonlinear feature interactions less effectively than the ensemble methods.

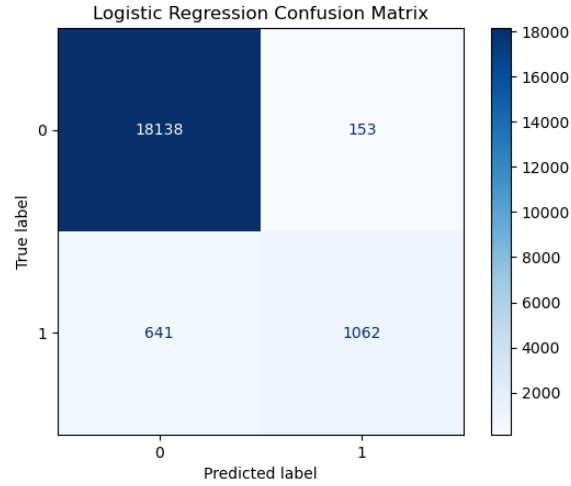


Fig. 10a — Logistic Regression confusion matrix.

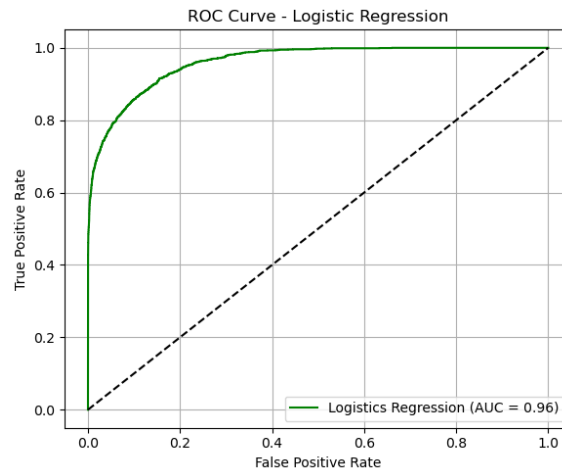


Fig. 10b — Logistic Regression ROC curve ($AUC = 0.96$).

4.4 K-Nearest Neighbors (KNN)

KNN achieved 0.96 accuracy, 0.90 precision, 0.57 recall, and 0.70 F1 score ($AUC = 0.89$). Its high precision but comparatively low recall indicates it misses a larger share of true diabetic cases than the ensemble models.

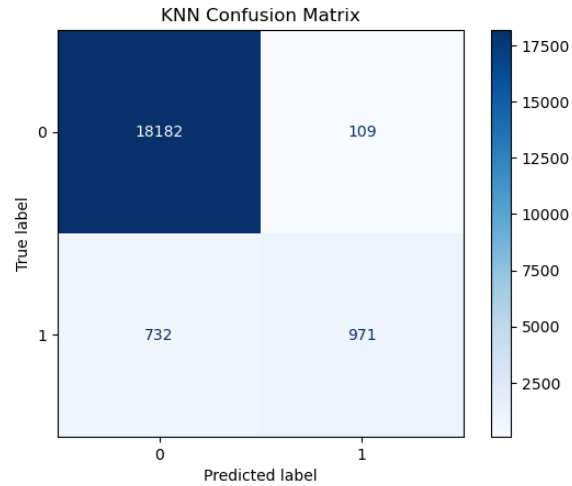


Fig. 11a — KNN confusion matrix.

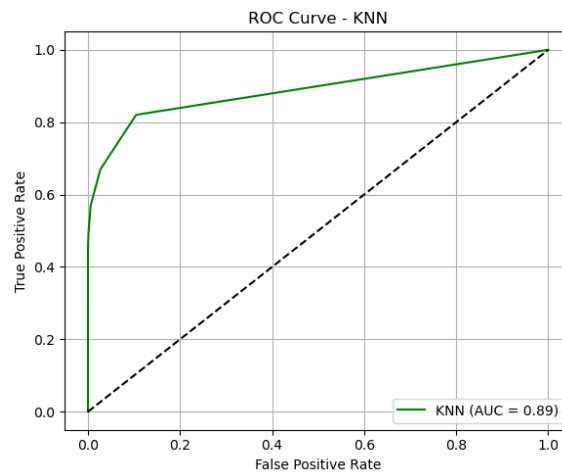


Fig. 11b — KNN ROC curve (AUC = 0.89).

4.5 Decision Tree

Decision Tree achieved 0.95 accuracy, 0.70 precision, 0.73 recall, and 0.71 F1 score (AUC = 0.85) — the highest recall of any single (non-ensemble) model, at the cost of more false positives.

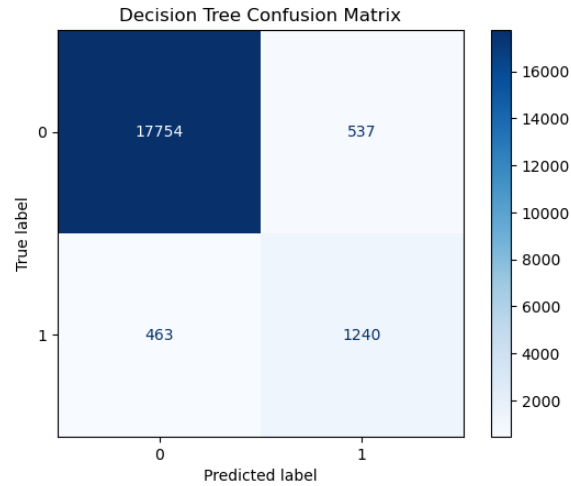


Fig. 12a — Decision Tree confusion matrix.

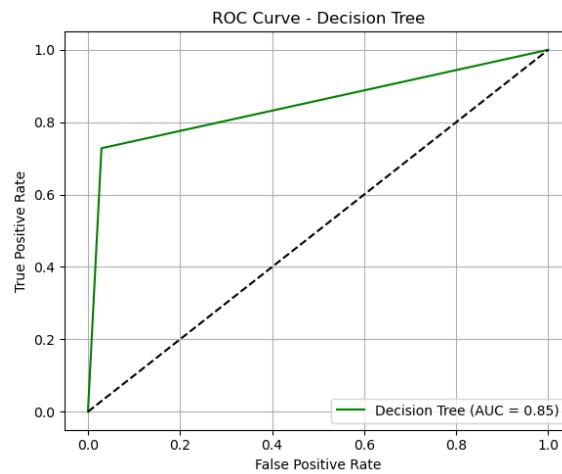


Fig. 12b — Decision Tree ROC curve (AUC = 0.85).

4.6 Naive Bayes

Naive Bayes showed the weakest performance of the six models: 0.91 accuracy, 0.46 precision, 0.66 recall, and 0.54 F1 score (AUC = 0.92). Its feature-independence assumption limits its ability to model the correlated clinical features in this dataset.

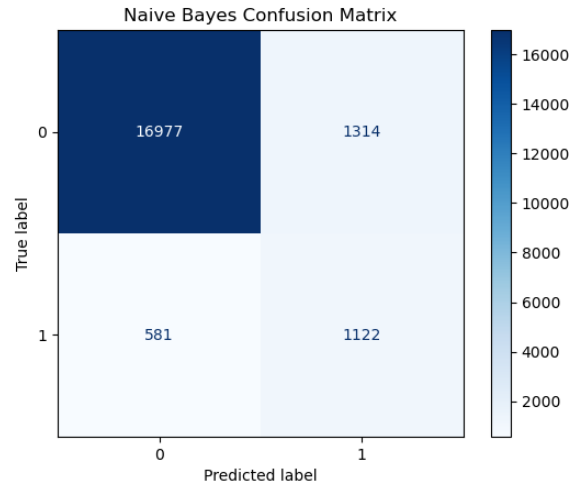


Fig. 13a — Naive Bayes confusion matrix.

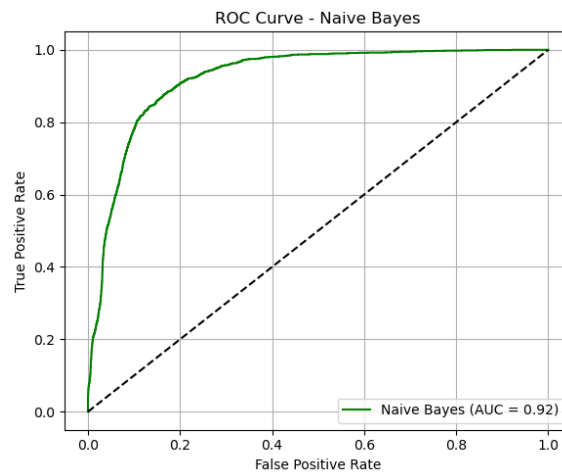
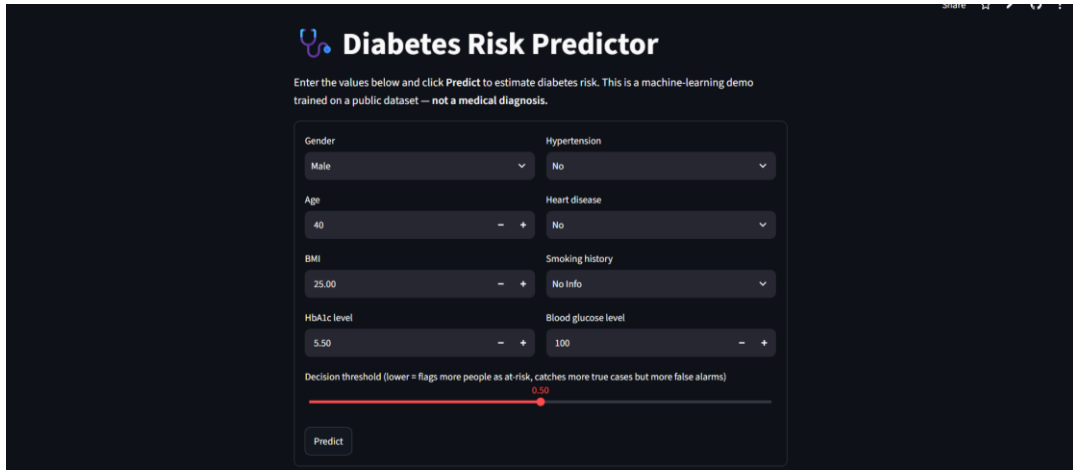


Fig. 13b — Naive Bayes ROC curve ($AUC = 0.92$).

5. Deployment

The trained model pipeline was deployed as an interactive web application, “Diabetes Risk Predictor,” built with Streamlit. The app accepts gender, age, hypertension, heart disease, BMI, smoking history, HbA1c level, and blood glucose level as inputs, plus an adjustable decision threshold, and returns an estimated diabetes risk probability. A clear on-screen disclaimer states that the tool is a machine-learning demo and not a medical diagnosis.



Diabetes Risk Predictor

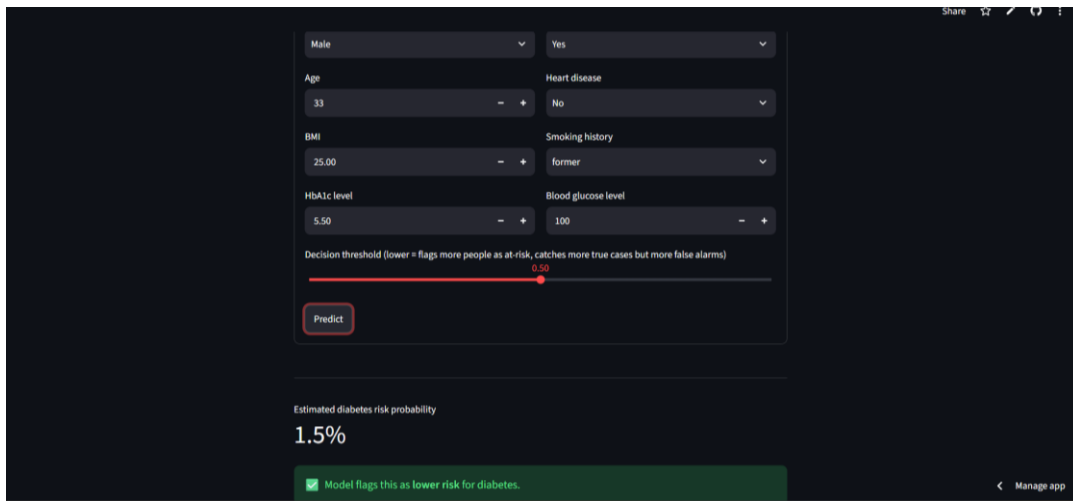
Enter the values below and click Predict to estimate diabetes risk. This is a machine-learning demo trained on a public dataset — not a medical diagnosis.

Gender Male	Hypertension No
Age 40	Heart disease No
BMI 25.00	Smoking history No info
HbA1c level 5.50	Blood glucose level 100

Decision threshold (lower = flags more people as at-risk, catches more true cases but more false alarms)
0.50

Predict

Fig. 14 — Diabetes Risk Predictor input interface: gender, age, hypertension, heart disease, BMI, smoking history, HbA1c level, blood glucose level, and decision threshold.



Male	Yes
Age 33	Heart disease No
BMI 25.00	Smoking history former
HbA1c level 5.50	Blood glucose level 100

Decision threshold (lower = flags more people as at-risk, catches more true cases but more false alarms)
0.50

Predict

Estimated diabetes risk probability
1.5%

☒ Model flags this as lower risk for diabetes.

Manage app

Fig. 15 — Example prediction: for a 33-year-old male with BMI 25, HbA1c 5.5%, and glucose 100 mg/dL, the app estimates a 1.5% diabetes risk probability and flags the case as lower risk.

6. Conclusion

This study confirms that HbA1c level and blood glucose level are the two dominant predictors of diabetes among the features examined, together accounting for roughly 70% of the Random Forest's feature importance — consistent with their established clinical significance.

Among the six models evaluated, XGBoost and Random Forest offered the best balance between precision and recall, achieving the highest F1 scores (0.81 and 0.80) and AUC scores (0.98 and 0.96). Logistic Regression and KNN performed respectably but trailed the ensemble methods, while Naive Bayes was the weakest performer due to its independence assumption. Because the dataset is heavily imbalanced (~91.5% non-diabetic), accuracy alone is a misleading metric here — F1 score and recall are far more informative for judging real diagnostic usefulness.

Future work could address the class imbalance more directly (e.g., SMOTE oversampling), tune ensemble hyperparameters further, validate on an external dataset, and extend the target to a multi-class formulation that distinguishes pre-diabetes from diabetes.

7. References

Breiman, L. (2001). Random forests. *Machine Learning*, 45(1), 5–32.
<https://doi.org/10.1023/a:1010933404324>

Diabetes Prediction Dataset — 100,000 patient records.
<https://www.kaggle.com/datasets/priyamchoksi/100000-diabetes-clinical-dataset/data>

ElSayed, N. A., et al. (2022). Introduction and Methodology: Standards of Care in Diabetes—2023. *Diabetes Care*, 46(Supplement_1), S1–S4. <https://doi.org/10.2337/dc23-sint>

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